AI Document Intelligence Report

Executive Summary

Overview

This sample report describes DocMind, a fictional document intelligence platform for portfolio demonstrations. The system ingests PDF documents, splits them into semantic chunks, embeds them with Hugging Face sentence-transformers, stores vectors in FAISS, and answers questions using retrieval-augmented generation (RAG) with a Groq-hosted large language model. The goal is accurate, evidence-backed answers with traceable sources rather than open-ended speculation.

Main Benefits

1. Grounded answers: responses are tied to retrieved passages from the uploaded PDF.
2. Low cost: embeddings run locally on CPU; inference uses Groq free-tier models.
3. Fast iteration: chunk size, retrieval k, and temperature are configurable via environment variables.
4. Portfolio-ready API: structured sources, confidence heuristics, and health endpoints.
5. Public demo safety: file size, page count, and question length limits protect shared hosting.

Limitations

The demo does not replace legal, medical, or financial advice. OCR quality depends on the source PDF. Very long documents may be truncated by page limits. Retrieval may miss relevant passages if chunk boundaries split critical sentences. The confidence score is heuristic, not a calibrated probability. Multi-document cross-referencing is limited in this single-index demo configuration.

Technologies Used

Backend: FastAPI and Uvicorn. RAG orchestration: LangChain community integrations. Embeddings: sentence-transformers/all-MiniLM-L6-v2 via langchain-huggingface. Vector store: FAISS (faiss-cpu). LLM: Groq ChatGroq with llama-3.3-70b-versatile default. PDF parsing: PyPDFLoader (pypdf). Deployment target: Hugging Face Spaces with Docker.

Suggested Demo Questions

What is this document about? What are the main benefits? What are the limitations? What technologies are used? Summarize the document.

Conclusion

DocMind illustrates how a small, free-tier stack can deliver credible document Q&A for portfolios and prototypes. Load this sample via POST /demo/load-sample to try the flow without uploading your own file.